

A chromomagnetic saddle amplifies the linearized flow at rate gB : sharpness of the truncation bound remains unproved

Later correction: the operator-inequality note supersedes the fixed-range competition in the transfer note; the lattice note separates a single lattice step from fixed-scale continuum estimates.

Correction (2026-10-02). The original title and lead claimed that the Nielsen–Olesen mode grows at $2gB$ and saturates the curvature-only Duhamel exponent. This omitted the lowest Landau energy gB : the full generator has growth rate $gB - k_{\parallel}^2$, at most gB . Symmetry alone also does not give a positive eigenvalue equal to the operator norm. The claimed lower bound on the spatial truncation error, the sharp crossover $\eta_* = 2g^2/c$, and the conclusion that large fields must be excluded from any flow argument are **withdrawn**. With $\sqrt{8t} = \ell$ and curvature norm η/ℓ^2 , the Duhamel exponent is $\eta/4$, not 2η . Sections 1–5 replace these claims in place; the symmetry proof and the existence of an unstable constant-field mode survive.

The curvature multiplication operator

$$(Mu)_{\mu} = 2[G_{\mu\nu}, u_{\nu}]$$

is symmetric, but the growth of the full linearized generator $L = D^2 + M$ includes the negative covariant-Laplacian contribution. In the constant abelian chromomagnetic example derived below, write $b = gB > 0$ for the curvature seen by a charge-one adjoint component. The lowest aligned Landau mode has

$$\omega^2 = k_{\parallel}^2 - b, \quad u(s) = e^{(b-k_{\parallel}^2)s} u(0).$$

Thus gradient descent can amplify perturbations around a saddle while lowering the action. Its maximum growth rate in this model is b , not $2b$. This proves an instability, **not** saturation of the upper bound e^{2bt} , a lower bound on its spatial tail, or an obstruction to all large-field flow estimates. No continuum construction or mass gap is proved here.

1. The curvature term is symmetric

Let $\mathfrak{g}$ be a compact matrix Lie algebra with invariant inner product $\langle X, Y \rangle = -\text{tr}(XY)$. Then $\text{ad}(Z)$ is antisymmetric: $\langle [Z, X], Y \rangle = -\langle X, [Z, Y] \rangle$.

Proposition 1. At each point, on $\mathfrak{g}$ -valued vectors with inner product $\sum_{\mu} \langle u_{\mu}, v_{\mu} \rangle$, the operator $(Mu)_{\mu} = 2[G_{\mu\nu}, u_{\nu}]$ is symmetric.

Proof.

$$\langle Mu, v \rangle = 2 \sum_{\mu\nu} \langle [G_{\mu\nu}, u_{\nu}], v_{\mu} \rangle = -2 \sum_{\mu\nu} \langle u_{\nu}, [G_{\mu\nu}, v_{\mu}] \rangle = 2 \sum_{\mu\nu} \langle u_{\nu}, [G_{\nu\mu}, v_{\mu}] \rangle = \langle u, Mv \rangle.$$

The two antisymmetries cancel. $\square$

A finite-dimensional symmetric operator of norm N has an eigenvalue of modulus N ; symmetry does not determine its sign. Moreover, positive eigenvalues of M alone do not determine those of $D^2 + M$. For compactly supported fields, integration by parts gives

$$\langle u, Lu \rangle = - \sum_{\alpha} \|D_{\alpha} u\|_2^2 + \langle u, Mu \rangle.$$

The first term must be retained.

For precision, let $\Gamma = \frac{1}{2} \sup_{s,x} \|M(s,x)\|_{\text{op}}$ on the flow interval. This is the curvature block-operator norm used in this note’s bounds; it is not an unspecified norm of individual tensor

components. For a covariant heat equation with bounded M , a free-heat-kernel majorant obtained by Duhamel has the form

$$|K_L(t; x, y)| \leq e^{2t\Gamma} K_t^{\text{free}}(x - y).$$

Indeed covariant diffusion is dominated in norm by scalar diffusion, and each iterated insertion of M is bounded by 2Γ ; scalar heat kernels compose, so the series sums to $e^{2t\Gamma}$. This is an **upper bound**, under those heat-equation hypotheses. It does not establish its own optimality. In the charge-one constant-field example below, $\|M\|_{\text{op}} = 2b$ and $\Gamma = b$.

2. The constant-field mode and its full growth rate

We work with the Yang–Mills flow linearized at a stationary background $\bar{A}$, in **background-covariant gauge**. This specifies the operator rather than assuming that every gauge-modified flow has the same Jacobian. With $D = \partial + [\bar{A}, \cdot]$ and $D_\nu G_{\nu\mu} = 0$, variation of the ungauged flow gives

$$\partial_s u_\mu = D^2 u_\mu - D_\mu D_\nu u_\nu + 2[G_{\mu\nu}, u_\nu].$$

Adding the linear background-gauge term $D_\mu D_\nu u_\nu$ gives L . Equivalently, on a mode with $D_\nu u_\nu = 0$, the ungauged linearization already agrees with L . This does not repair or identify the full Jacobian in every gauge convention of the Jacobian note.

Choose an $SU(2)$ colour sector with $[T^3, E_+] = iE_+$, and absorb the coupling into the connection, so $G_{12} = bT^3$, $b = gB > 0$. In symmetric gauge on the transverse plane,

$$\bar{A}_1 = -\frac{b}{2}x_2T^3, \quad \bar{A}_2 = \frac{b}{2}x_1T^3.$$

On the E_+ component,

$$D_1 = \partial_1 - \frac{ibx_2}{2}, \quad D_2 = \partial_2 + \frac{ibx_1}{2}.$$

Writing $Q = D_1 - iD_2$, integration by parts and $[D_1, D_2] = ib$ give $Q^*Q = -(D_1^2 + D_2^2) - b$ and $[Q, Q^*] = 2b$. Thus the oscillator ladder has transverse magnetic-Laplacian energies $(2n + 1)b$, $n = 0, 1, \dots$. To fix the factor without relying on a bibliographic convention, its lowest mode can be checked directly:

$$f(x_1, x_2) = e^{-b(x_1^2 + x_2^2)/4}, \quad (D_1 - iD_2)f = 0, \quad (D_1^2 + D_2^2)f = -bf.$$

On the two transverse vector components,

$$M = \begin{pmatrix} 0 & 2ib \\ -2ib & 0 \end{pmatrix}, \quad M \begin{pmatrix} f \\ -if \end{pmatrix} = 2b \begin{pmatrix} f \\ -if \end{pmatrix}.$$

The charged complex notation represents real Lie-algebra fields by adding the conjugate charge sector. The vector

$$u = (f, -if, 0, \dots)e^{ik_\parallel \cdot x_\parallel}$$

is background-transverse, since $D_1 f - iD_2 f = 0$. Consequently

$$D^2 u = -(b + k_\parallel^2)u, \quad Lu = (b - k_\parallel^2)u.$$

This supplies both the positive eigenvalue of M and the Laplacian cost on the same vector. The two transverse polarizations at general Landau level have the spectral values

$$\omega^2 = k_\parallel^2 + (2n + 1)b \mp 2b$$

for $-L$. Here $k_{\parallel}^2$ is the sum of the squared momenta in directions outside the magnetic plane: one for spatial three-dimensional flow, two for four-dimensional Euclidean flow. The unstable branch is

$$\omega^2 = k_{\parallel}^2 - b < 0 \quad \text{when } k_{\parallel}^2 < b,$$

and its gradient-flow growth is

$$e^{-\omega^2 s} = e^{(b-k_{\parallel}^2)s}.$$

At $k_{\parallel} = 0$ the rate is b , half the exponent coefficient $2b$ in the Duhamel majorant. That example therefore **does not saturate** the majorant. Whether a better universal bound holds for other backgrounds is not settled by this calculation.

On the whole space the longitudinal plane waves are generalized modes; wave packets supported in $k_{\parallel}^2 < b$ give growing L^2 perturbations, with rates approaching b . Periodic longitudinal boundary conditions admit the zero-momentum mode. The constant field on the transverse plane has infinite total action; the calculation is a stationary-background spectral model, not a finite-action four-dimensional construction or a statement for arbitrary finite boxes.

This is the instability historically associated with Nielsen and Olesen, “An unstable Yang–Mills field mode,” Nucl. Phys. B144 (1978) 376–396 (abstract read; the original spectral passage was not read here). The numerical factors used above are supported by the displayed calculation, not by that abstract or a normalization left implicit.

3. Why growth is consistent with action monotonicity

For the unmodified gradient flow with its gradient-metric normalization,

$$\frac{d}{ds} S(A_s) = -\|\nabla S(A_s)\|_2^2 \leq 0.$$

A constant abelian curvature has $D_{\nu}G_{\nu\mu} = 0$ and is stationary. Its transverse unstable modes give negative Hessian directions, $\text{Hess } S = -L$ on the background-transverse sector in the normalization of Section 2. Nearby trajectories can separate at rate $b - k_{\parallel}^2$ while descending the action. For the infinite-action background these are statements about the local equation and quadratic variation on admissible perturbations, rather than differentiation of an infinite total action.

The original comparison with the valley note is also corrected: constant commuting connections in its flat holonomy valley have $G = 0$. They are not the nonzero constant chromomagnetic background considered here. The valley’s zero-point potential is a different calculation and gives no justification for this saddle or its growth rate.

4. Consequence: instability does not determine the truncation tail

The Gaussian majorant in Section 1 gives a bound on the **Jacobian kernel tail**. Writing $R = \kappa\sqrt{8t}$ in three spatial dimensions,

$$\int_{|x-y|>R} |K_L(t; x, y)| dy \leq C(1 + R/\sqrt{t}) \exp[2t\Gamma - 2\kappa^2].$$

This is an upper bound with its polynomial prefactor retained. Converting it into a relative form bound for the conjugated Hamiltonian of the flow-conjugation note requires additional operator estimates; this note proves no such conversion. A large upper bound does not imply a large actual error. Nor would sharpness of a global semigroup norm imply sharpness of a discarded spatial tail: the lowest Landau function above is localized on transverse length $b^{-1/2}$. A tail lower bound would require a separate spatial calculation.

The corrected scale and range implications are:

- At $\ell = \sqrt{8t}$ and $\Gamma = \eta/\ell^2$, the bound's growth factor is $e^{\eta/4}$, since $2t\Gamma = \eta/4$; the example's zero-momentum growth is $e^{bt} = e^{\eta/8}$ when $b = \eta/\ell^2$. If $\Gamma \leq \eta/\ell^2$, the former is an upper bound. A lower threshold $\Gamma \geq \eta/\ell^2$ gives no lower bound on the tail.
- The earlier comparison of $e^{-c\eta^2/g^2}$ with $e^{2\eta}$ and its claimed sharp crossover $2g^2/c$ are withdrawn. Even if one **assumes** a normalized probability estimate $e^{-c\eta^2/g^2}$, multiplying it by the corrected growth majorant yields only the schematic exponent $-c\eta^2/g^2 + \eta/4$ at fixed range, with algebraic zero $g^2/(4c)$. This is bookkeeping of assumed upper bounds, not a physical threshold or a sharp truncation theorem; an action cost alone does not prove the assumed probability bound.
- As already noted in the transfer note and the operator-inequality note, the fixed-range competition is superseded in part. For a finite curvature upper bound, increasing κ makes $2t\Gamma - 2\kappa^2$ negative enough to control the kernel tail, including its prefactor. Whether the resulting range and operator errors permit an iterated construction is a further obligation.

What this settles. The constant-field example rules out a universal contraction claim for the linearized flow in this sector. It leaves open improved large-field kernel bounds, Hamiltonian truncation bounds and iteration. The earlier conclusion that every large-field region must be excluded is withdrawn. The lattice note separately discusses bounded coefficients over one lattice doubling; its conclusions must not be inferred from this continuum saddle calculation.

5. Consequence for STATE

The surviving result is the symmetry of M and a stationary-background instability with growth $gB - k_{\parallel}^2$, compatible with gradient descent. This does not complete the flow-based route on large fields: sharpness of the curvature-only exponent and a lower bound on the truncation error remain unproved. The fixed-range crossover and the necessity of excluding large fields are withdrawn. The relevant open estimate is a spatial-tail and operator comparison with explicit curvature, range and lattice-step dependence, together with whatever normalized measure control its use requires. A convergent large-field expansion and the continuum $SU(3)$ mass gap are not consequences of this note.